

OOP

programare orientată pe obiecte

CURS10: FRAMEWORKS (INVERSION OF CONTROL, DEPENDENCY INJECTION, PATTERNS)

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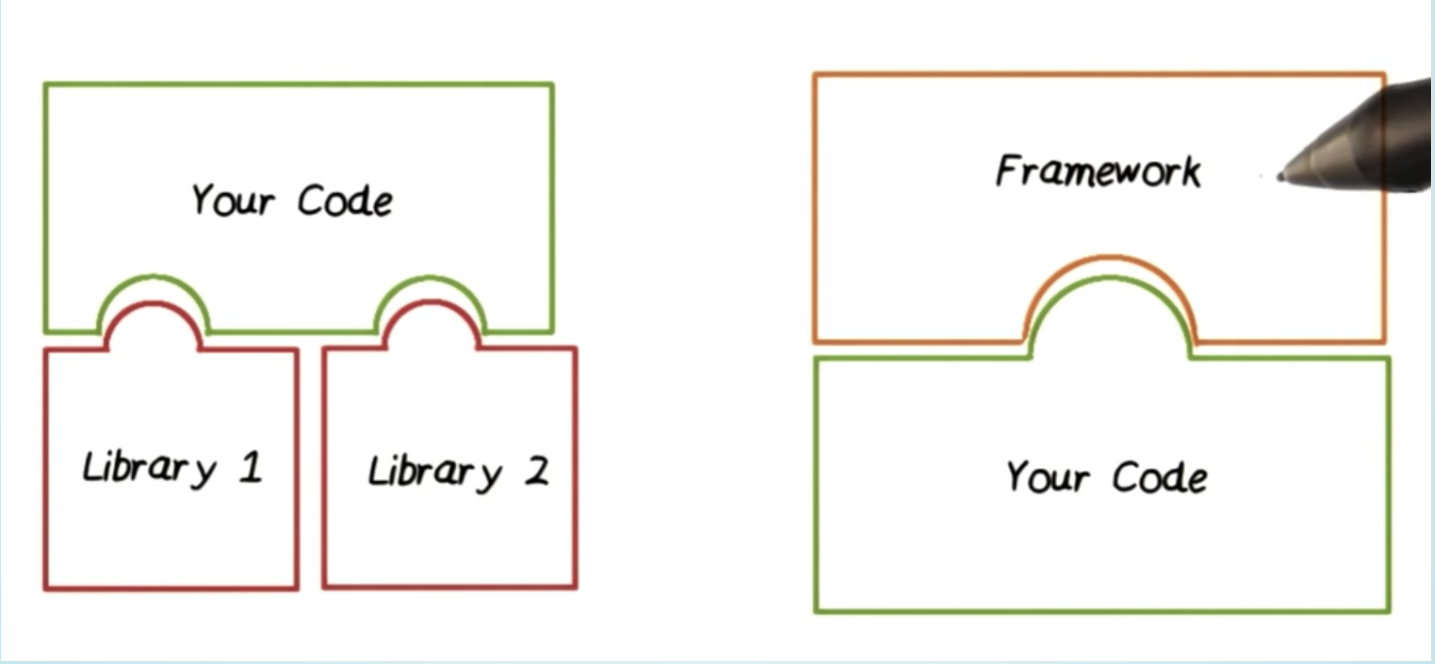




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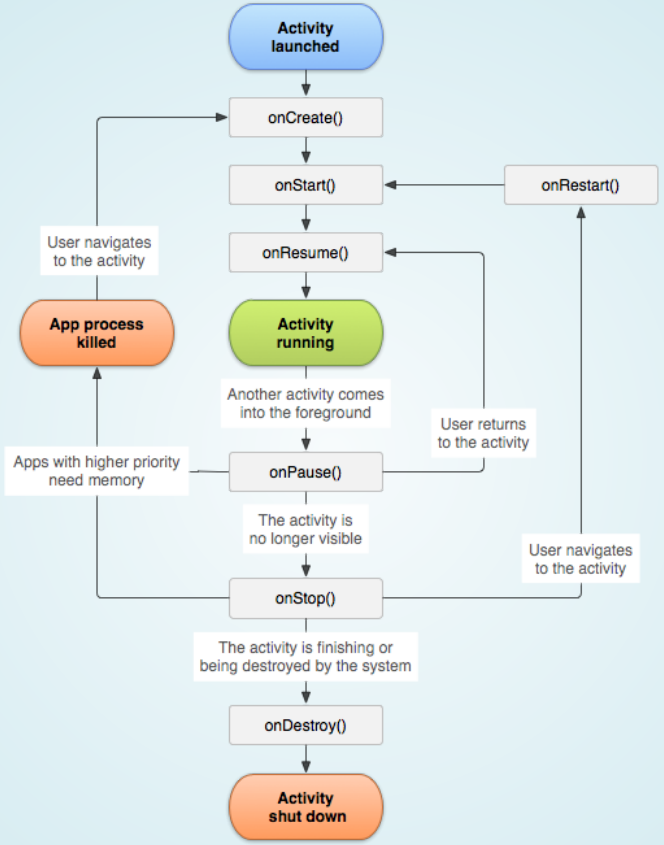
A software framework is an abstraction in which software providing generic functionality can be selectively changed by additional user-written code, thus providing application-specific software

Inversion of Control is a design principle in which custom code you write receives flow of control from a generic framework (as opposed to a 'traditional' architecture where custom code you write is the one that calls into reusable libraries)



Inversion of Control Youtube video

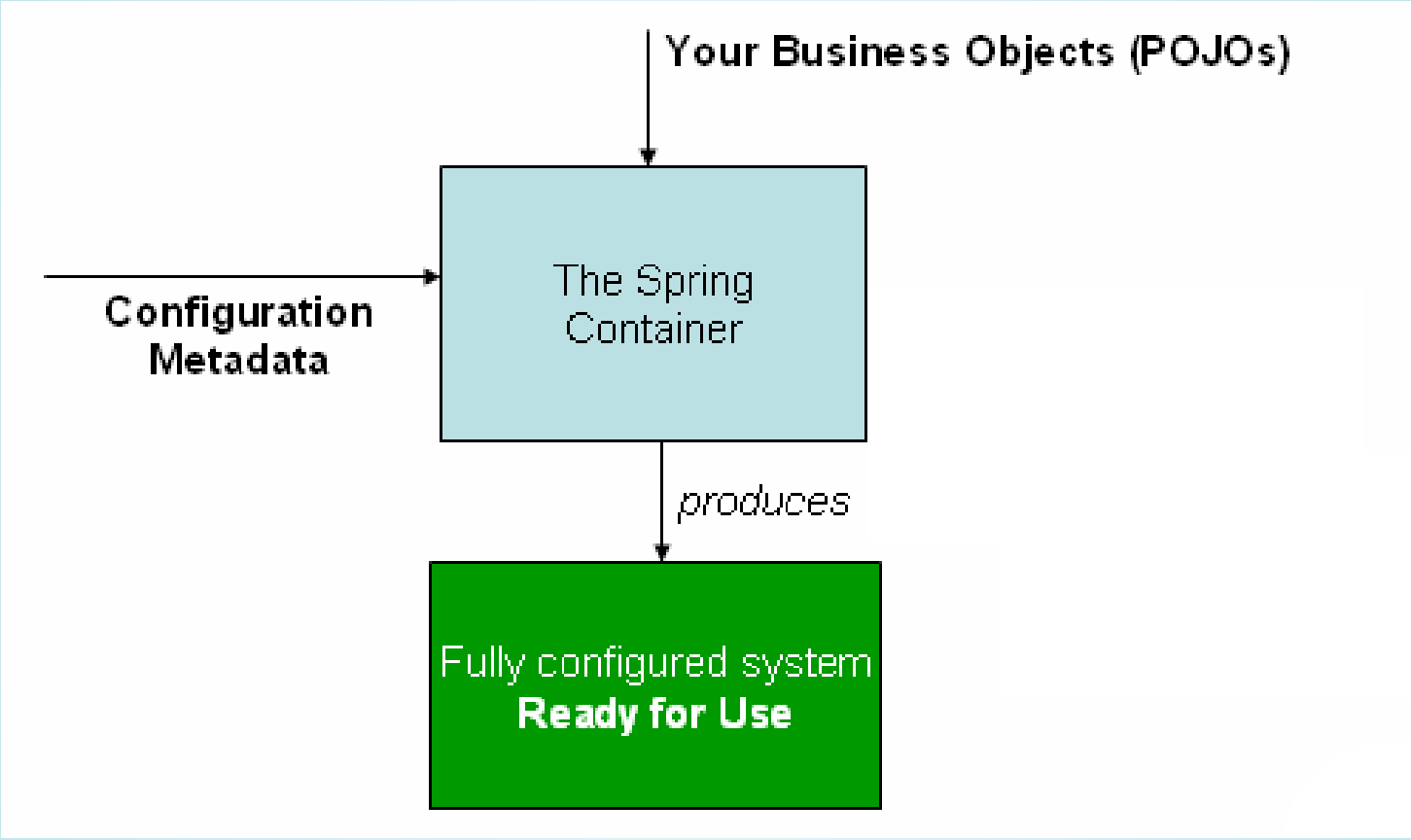
Android Studio > New Project > Basic Activity



Activity Lifecycle Callbacks

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IntellijIDEA > File > New > Project > Spring Initializr



Spring Docs: The IoC Container

Many implementation techniques, relying heavily on design patterns:

- Service Locator pattern
- Dependency Injection
- Contextualized lookup
- Template Method pattern
- Strategy pattern

Dependency Injection is a design pattern that:

- requires custom classes to link to Dependencies through setters or constructors (instead of instantiating with new)
- allows frameworks to inject proper implementations to those Dependencies (aka autowiring in some frameworks)

Dependency Injection vs Dependency Inversion?

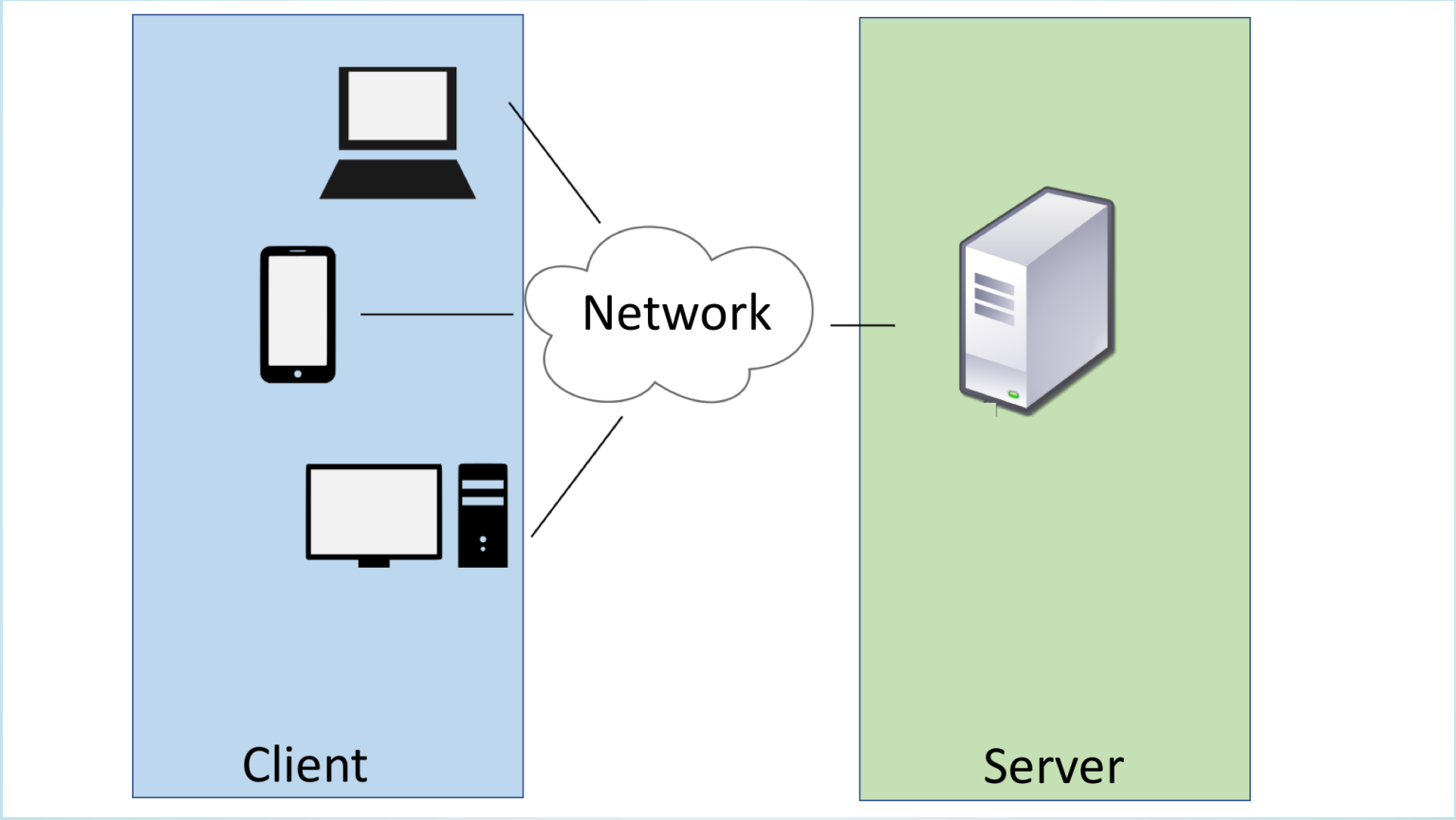
A Spring bean is basically an object managed by Spring. It is an object that is instantiated, configured and otherwise managed by a Spring Framework container. Spring beans are defined in Spring configuration files (or, more recently, with annotations), instantiated by Spring containers, and then injected into applications.

Demo app Spring

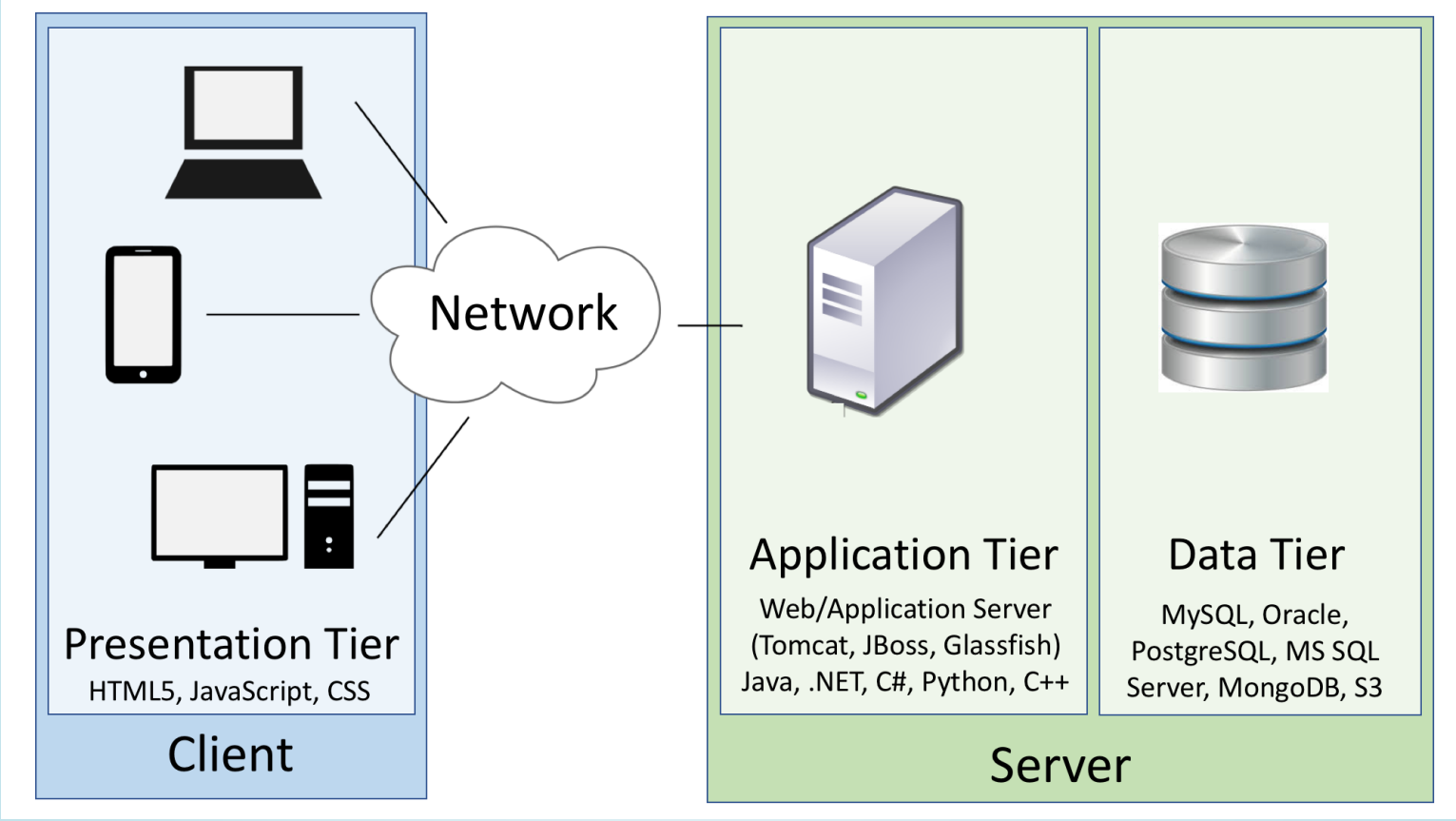
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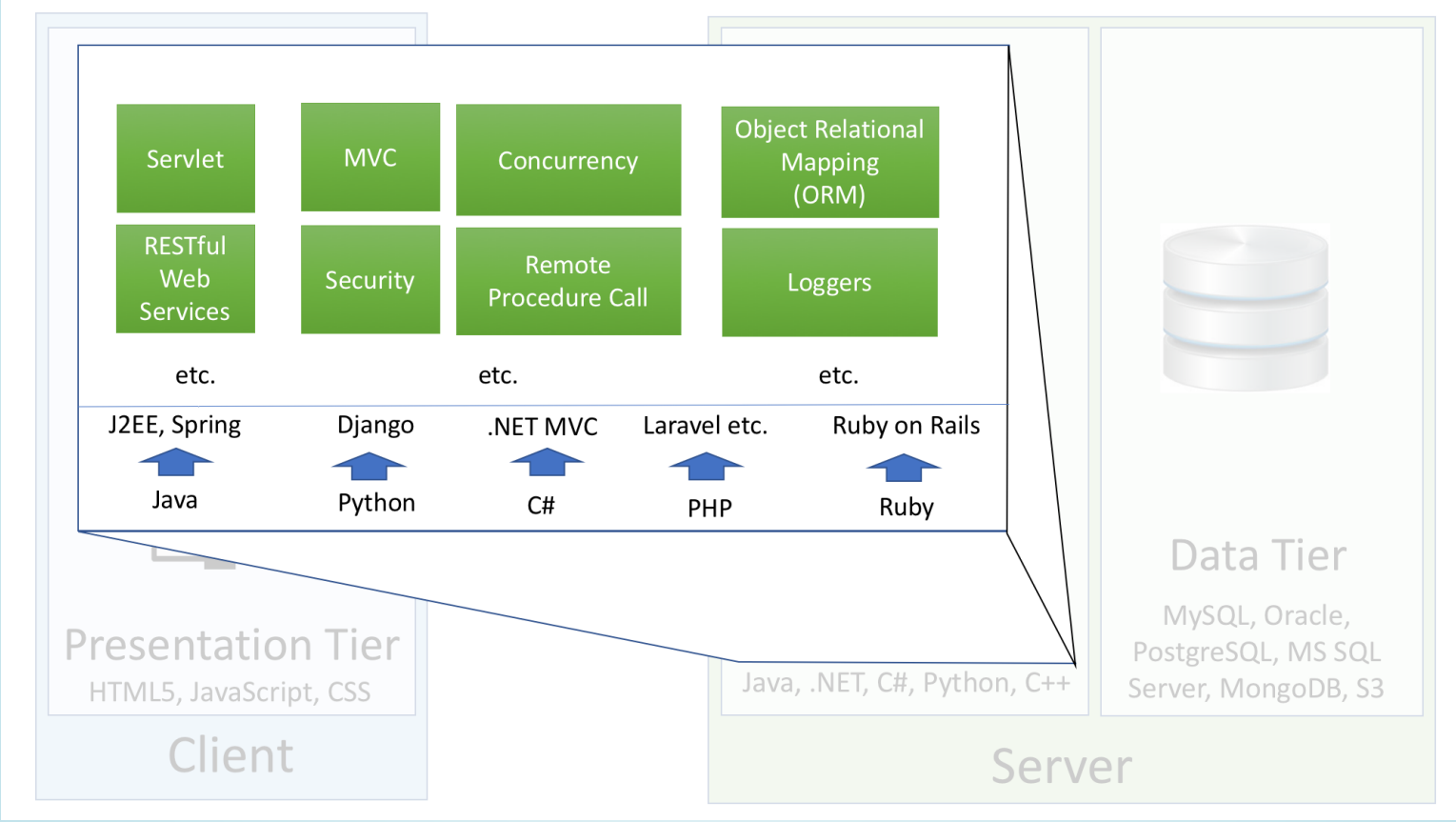
Multi-tier architectures



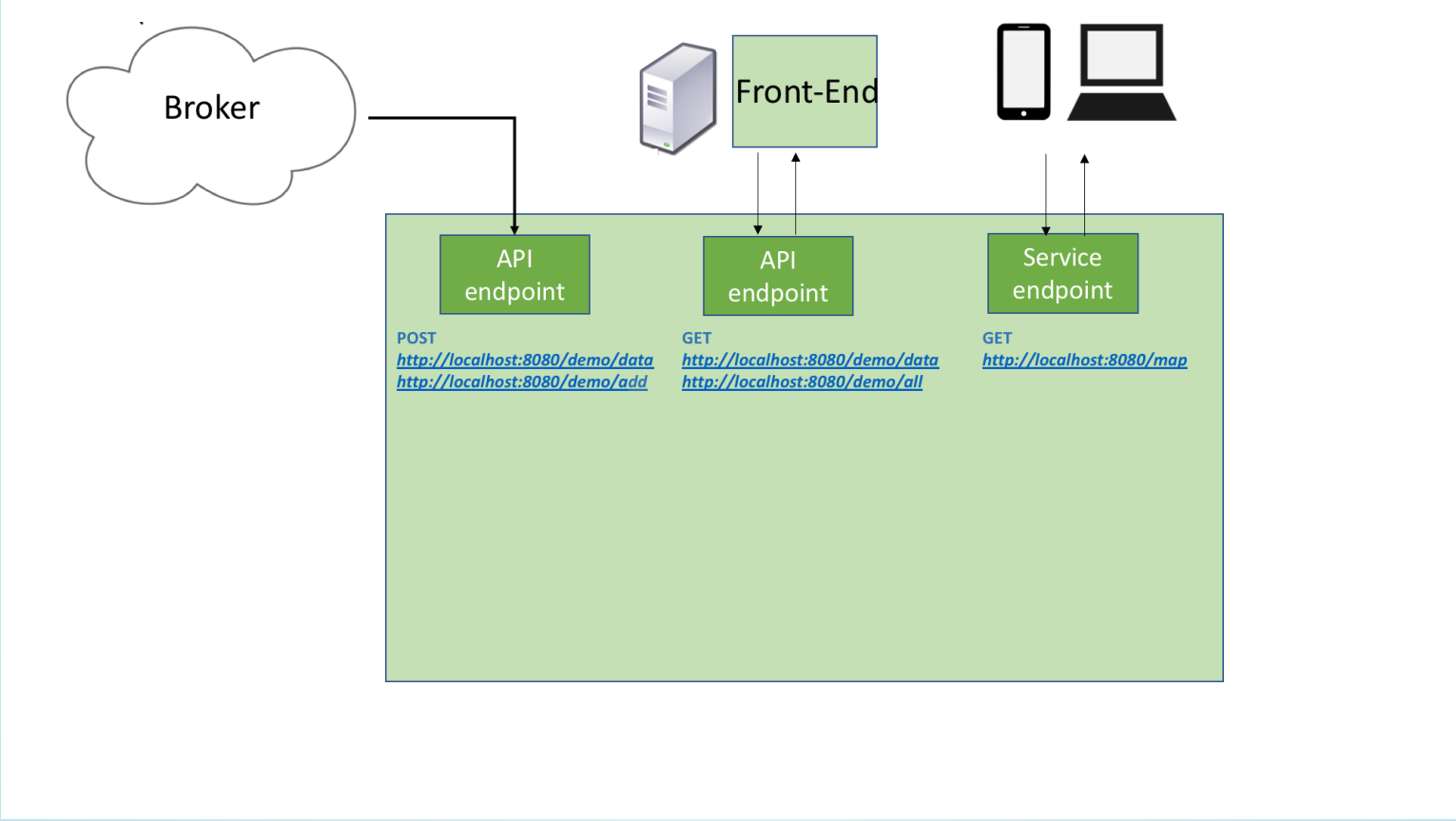
Multiple possible technologies in each tier



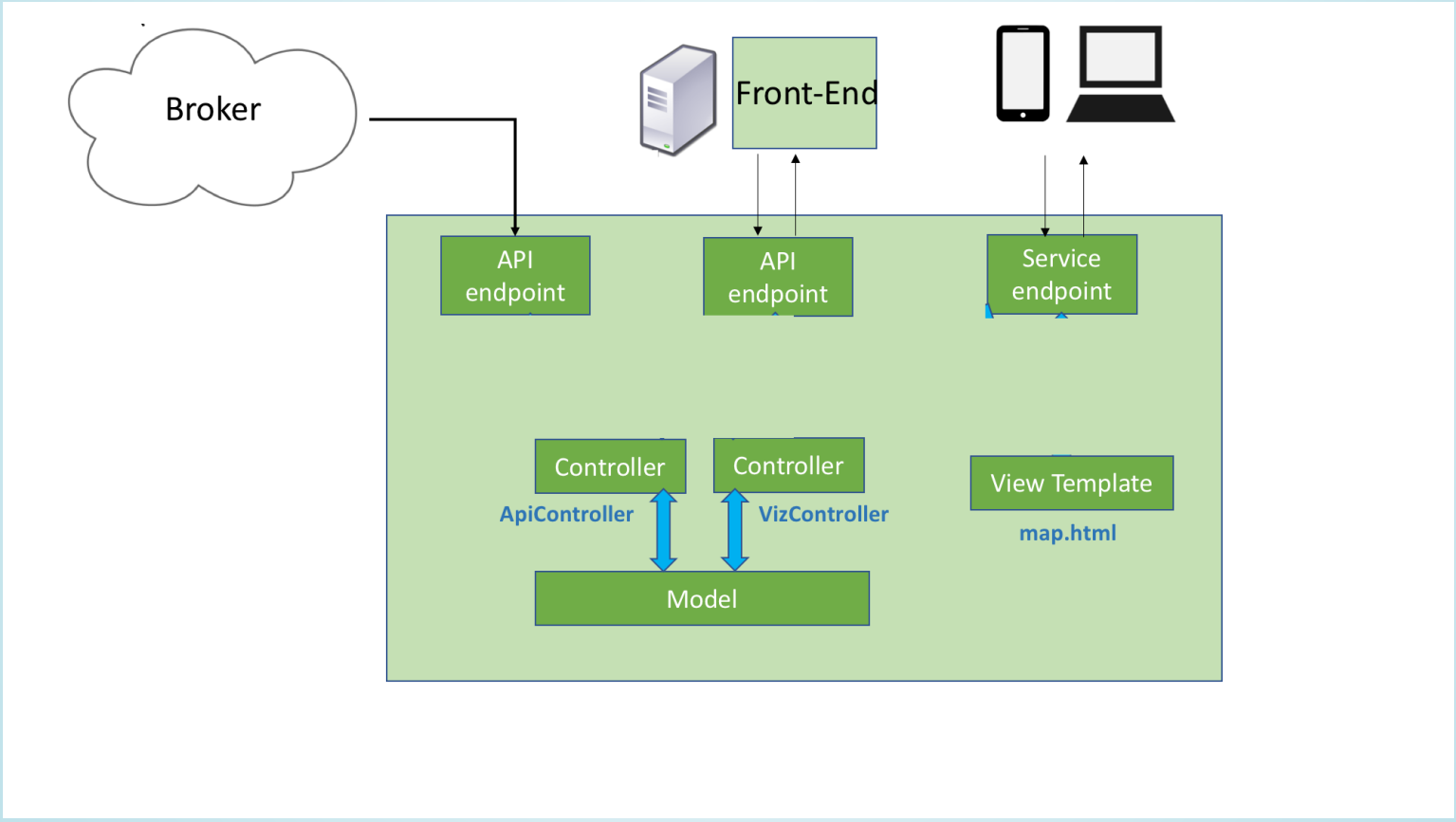
Frameworks in the application layer



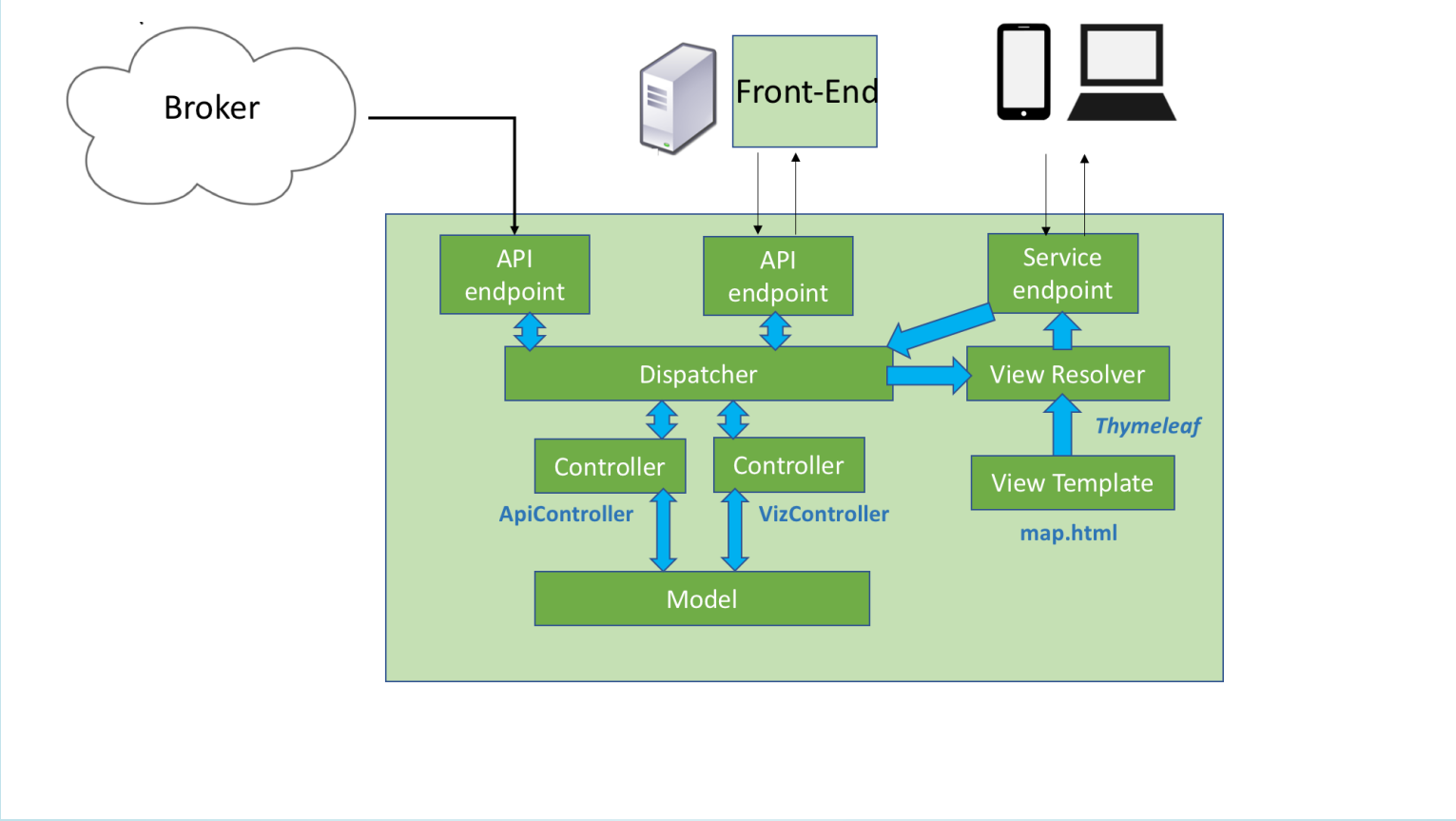
The application is a blackbox exposing endpoints



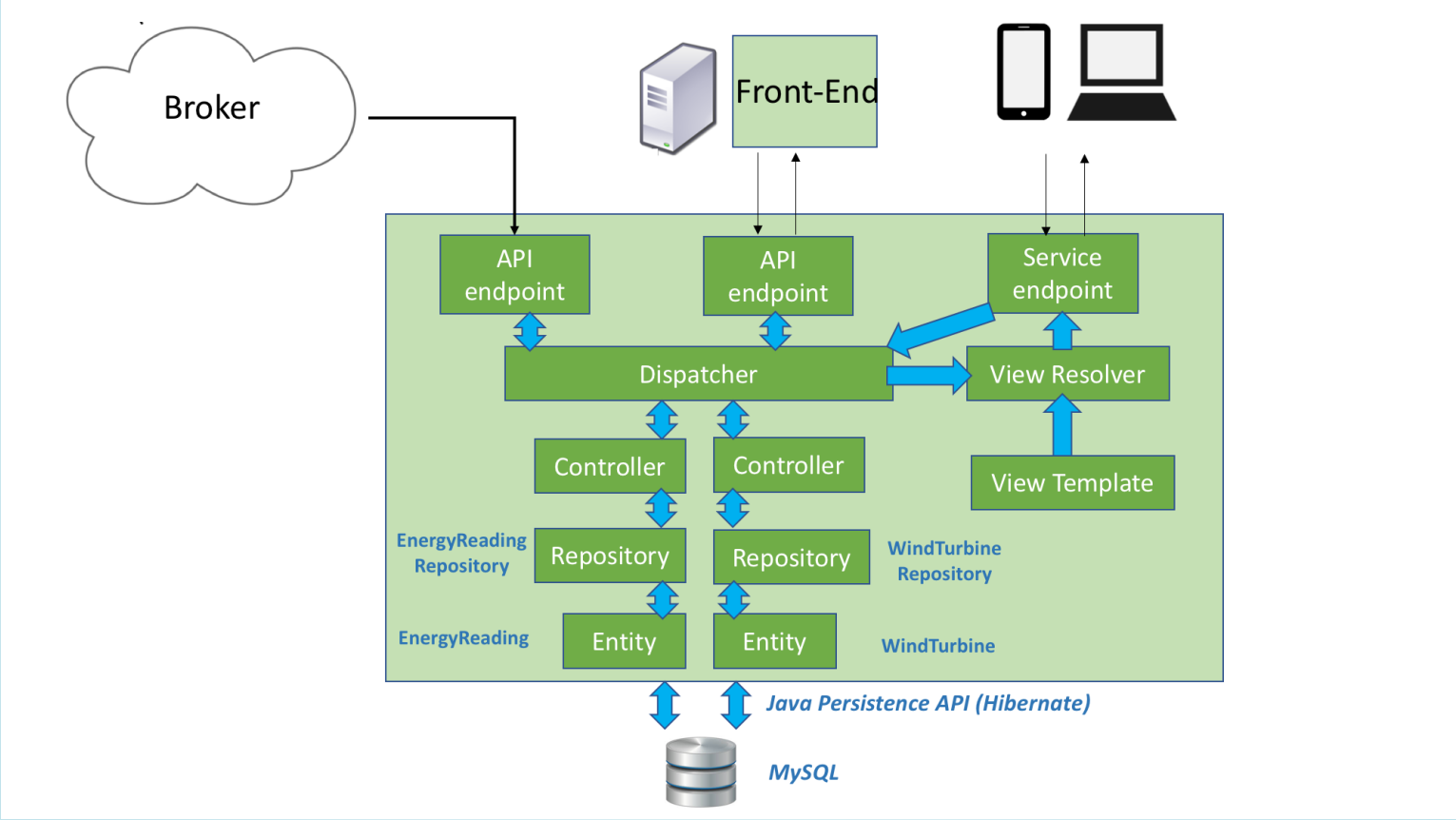
Inside, typically code follows an architectural pattern (MVC, MVP, MVVM etc)



The connection between MVC and the endpoints



The connection between MVC and the data tier



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Reading material:

Difference between dependency injection and dependency inversion

MVC vs. MVP vs. MVVM, Niraj Bhatt

Repository Pattern - A data persistence abstraction

Why shouldn't I use the Repository Pattern with Entity Framework